

ShadeMate — B2B Project Blueprint

Confirmed Architecture & Strategy Document

Version 1.2 — Phase 1 Only

1. What ShadeMate Is

ShadeMate is a **B2B SaaS product** — an AI-powered shade recommendation widget that plugs into existing beauty e-commerce platforms.

It is **not** a standalone app or e-commerce store.

It is a **service sold to small and mid-scale beauty businesses** that already have their own website or Shopify store.

Target clients: Any small/mid-scale beauty e-commerce business with their own online store — on Shopify, WooCommerce, custom websites, or any other platform. Not marketplace-only sellers.

One-line pitch:

"We reduce return rates for beauty e-commerce businesses by helping their customers find the right shade before they buy — using AI skin tone detection."

2. The Problem

- Customers buying colour cosmetics online (foundation, concealer, lipstick, blush) order multiple shades to try at home
- They keep the one that suits them and return the rest
- This is not fraud — it is a **shade-bracketing behavior** driven by zero guidance at point of purchase
- Small and mid-scale beauty businesses absorb these return losses without any technical solution
- Large players (Nykaa, Myntra) build this in-house — small businesses cannot

Result for small businesses:

- Revenue loss on every returned order
- Reverse logistics alone costs 25–30% of order value on sub-₹1,500 items
- Inventory blocked by returned stock
- Lower customer retention and trust
- No technical solution they can afford or implement themselves

Important framing note — do NOT cite a generic beauty return rate. The problem is specifically **shade-bracketing behavior** in colour cosmetics. There is no single clean India-specific stat for this. Frame it as an observed, relatable behavior — not a percentage. Judges who push back on a made-up stat will derail the entire pitch.

3. The Solution

A **plug-and-play AI widget** embedded on the client's existing product pages.

When a customer visits a foundation or lipstick page, the widget appears as a "**Find Your Shade**" button.

The widget:

1. Takes a customer selfie
 2. Checks lighting quality before proceeding
 3. Detects their skin tone from 3 facial regions
 4. Matches against THIS store's product catalogue
 5. Recommends the top 3-5 best-fit shades from that store's inventory
 6. Customer adds the right product directly to cart — first time, every time
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4. Target Customers (Our Clients)

- Small to mid-scale beauty e-commerce businesses in India
- Selling via their **own website** — Shopify, WooCommerce, or custom built
- Do **not** have an in-house development team
- Cannot build this themselves
- Revenue range: ₹2L - ₹50L/month
- Examples: indie makeup brands, D2C skincare stores, multi-brand beauty retailers, regional cosmetics brands

End user vs Paying customer distinction:

- **End user** = the beauty store's customer (who takes the selfie and gets the recommendation)
- **Paying customer** = the beauty store owner (who pays ShadeMate monthly)

Why they buy:

ROI is immediately calculable.

"If we prevent 10 returns/month at ₹800 average order value = ₹8,000 saved against ₹2,000 tool cost. 4x positive from Day 1."

5. Phase Roadmap

Phase 1 — Beauty Only (Current Build)

- Skin tone detection from selfie
- Shade matching against beauty product catalogue
- Widget embed on beauty product pages
- Shopify + script tag integration for any website
- Supported products: Foundation, Concealer, Lipstick, Blush, Eyeshadow

Phase 2 — Clothing + AR Added

- Color harmony engine for clothing/apparel
- Separate widget mode for clothing stores
- Occasion selector (Daily / Office / Wedding / Festival)
- AR face overlay for virtual try-on of lip and eye products

Phase 3 — Combination + Occasion Intelligence

- Full outfit + makeup combination recommendations
 - Cross-sell between beauty and clothing on same platform
 - Occasion-based filtering fully integrated
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6. How Integration Works

For Clients

- Client signs up on ShadeMate merchant portal
- Connects their Shopify store via **one-click Shopify OAuth** (wizard-guided, no technical knowledge needed)
- OR for non-Shopify stores: manually uploads product catalogue CSV
- Product catalogue auto-pulled via Shopify Admin API
- Client fills in shade details per product (onboarding form — see Section 8)
- Client receives **one script tag** to paste in their website

```
<script src="https://shademate.in/widget.js" data-key="YOUR_API_KEY"></script>
```

- Widget goes live on all product pages immediately
- **ShadeMate onboarding team assists every new client with setup** — no client is left alone during integration

API Key Security

- Each API key is **domain-locked** to the client's registered store URL
- Requests from unregistered domains are rejected at FastAPI level
- Prevents key theft and unauthorized usage

7. Onboarding Form — Per Product Data Collection

When client's catalogue is synced from Shopify, they fill in the following per product:

Field	Input Type	Source
Product Name	Text (auto-filled)	Shopify / CSV
Product Image	Image (auto-filled)	Shopify / CSV
Product Type	Dropdown: Foundation / Concealer / Lipstick / Blush / Eyeshadow	Client
Shade Color	Predefined shade palette picker (~50 standardized shades)	Client
Undertone	Dropdown: Warm / Cool / Neutral	Client
Finish	Dropdown: Matte / Dewy / Satin	Client
Price	Auto-filled	Shopify / CSV

Shade Color Input — Important Detail

- Client is **not asked for a hex code** (they won't know it)
- Instead they select from a **predefined palette of ~50 standardized shades** covering the full Indian beauty spectrum (fair to deep)
- Palette is screen-calibration independent — avoids monitor color variance issues
- Advanced free-form color picker available optionally for precise clients
- Selected palette color becomes the **anchor point** for that product's search space
- System defines search space = all shades within **Delta-E ≤ 12** of the anchor

8. The Recommendation Algorithm

Step 1 — Selfie Quality Check

- Widget runs a **brightness/lighting quality check** using Canvas API in browser
- If too dark or too bright → prompts: *"Please move to better lighting"*
- Only confirmed quality images proceed to the model

Step 2 — Skin Tone Extraction

- Face detected using **MediaPipe Face Mesh**
- Skin sampled from **3 regions**: forehead, left cheek, right cheek
- Average of 3 regions taken → converted to **LAB color space**
- Undertone detected (warm/cool/neutral) from skin color profile

Step 3 — Search Space Definition

- Each product has a client-picked anchor color from the predefined palette
- Matching only happens within **Delta-E ≤ 12** of that anchor
- Client input narrows the search intelligently — reduces noise, improves accuracy

Step 4 — Product Type Routing

- **Foundation / Concealer** → **analogous/matching** shade logic (find shades matching skin tone)
- **Lipstick / Blush / Eyeshadow** → **complementary** shade logic (find shades complementing skin tone)
- Product type tag collected during onboarding determines which logic runs

Step 5 — Delta-E Scoring

- Calculate **Delta-E distance** between product shade and customer's detected skin tone
- Secondary filter: **undertone match** (warm/cool/neutral must align)
- Products passing both get a combined score

Step 6 — Threshold Filter & Output

- Only products **above threshold score** returned
- If zero products pass → threshold **relaxed by 20% and rerun once** (guarantees results)
- Top 3-5 results returned, ranked by score

- Labels: "Best Pick" / "Great Match" — never negative framing
 - Confidence scores stay **internal** — never shown to end customer
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9. Widget Output Design

What the Customer Sees After Recommendation



- Two colored circles side by side — detected skin tone + recommended product shade
- Product name, shade name, and match label clearly shown
- Direct "Add to Cart" link
- Top 3-5 cards shown, ranked by score

Why Not AR / Virtual Try-On in Phase 1

- Requires facial landmark mapping + realistic shader blending
- Weeks of development beyond current team bandwidth
- Core value is **accurate shade matching**, not visual simulation
- AR face try-on is Phase 2 roadmap item

Pitch Line for This

"Phase 1 gives customers a precise color match comparison — their skin tone alongside the recommended shade — delivering purchase confidence through color science. Virtual face try-on using facial AR is our Phase 2 feature."

10. Tech Stack

Component	Technology	Owner
AI/ML Service	Python + FastAPI + OpenCV + MediaPipe	Milind
Skin Tone Model	LAB color space + Delta-E scoring	Milind
Backend API	Spring Boot + PostgreSQL	Sujal
JS Embed Widget	React + Tailwind (compiled to single JS file)	Uvaish
Merchant Dashboard	React + Vercel	Uvaish
Mobile Demo App	Flutter + Firebase	Shreya
Hosting (AI Service)	Render.com / Railway.app (free tier)	Milind
Design	Figma	Nikul + Harshil + Sahil

Why This Stack Works

- FastAPI is lightweight and handles image processing efficiently
- MediaPipe runs server-side without GPU requirement
- React widget compiles to a single JS file — embeds anywhere
- Shopify Admin API is REST-based — straightforward to call from FastAPI
- Flutter demo shows the mobile customer experience end-to-end

Cost

- **Zero mandatory spend** to build and demo this product
- All tools and hosting are free/open-source at demo scale
- Optional: domain name ~₹500-800/year for a `.in` domain

11. Data & Privacy

- **Zero photo storage** — customer selfie processed in-memory, immediately discarded
- Only anonymized tone metadata (LAB values) used for matching
- No personally identifiable information stored
- ShadeMate operates as a **DPDP-compliant data processor**
- Data Processing Agreement signed with every client at onboarding

- API keys are domain-locked — no cross-client data leakage

12. Market Opportunity

The Big Picture

- India's beauty and personal care market: **\$28 billion**, growing at 10-11% annually (IBEF 2025)
- India's online beauty sales grew **39% in just 6 months** in 2024 — world's fastest growing online beauty market (NielsenIQ via IBEF)
- India's D2C beauty market grew **88% in 2024**, reaching \$2.1 billion (StartupIndiaX)
- 53% of Indian MSMEs now prefer selling via their own D2C store (McKinsey 2025)

Your Serviceable Market

- 93,700 Shopify stores in India (April 2025), ~11% in Beauty & Fitness = **~10,300 Shopify beauty stores**
- WooCommerce has 2x+ Shopify's India market share = **~20,000+ WooCommerce beauty stores**
- Custom/other platforms = **~5,000+ additional**
- **Total addressable universe: ~35,000 small/mid beauty e-commerce businesses with their own website**
- 95%+ run by teams of fewer than 10 people — no technical team to build this themselves

Realistic Market Sizing (Year 1 → Year 3)

Stage	Metric	Number
Total addressable	Small/mid beauty stores with own website	~35,000
Year 1 reachable	2% of market	~700 businesses
Year 1 paid conversion	30% of reachable	~210 clients
Year 1 monthly revenue	210 × ₹2,500 avg	~₹5.25 lakh/month
Year 1 annual revenue		~₹63 lakh
Year 3 paid clients	5% of total market	~1,750 clients
Year 3 annual revenue	1,750 × ₹2,500/month	~₹5.25 crore

The D2C Timing Argument

- 166+ D2C beauty startups in India as of 2026, \$800M+ in VC funding (Inventiva)
 - These brands all have their own websites — and none have shade intelligence built in
 - They are our perfect early adopter — digitally savvy, growth-focused, no dev team
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13. Business Model

Pricing Model — Results-First

Entry: Zero upfront cost. Widget goes live, we handle full setup.

60-Day Results Window: Widget runs for 60 days collecting real data:

- Recommendations served
- Estimated returns avoided (recommendations × conversion benchmark)
- Estimated revenue saved (returns avoided × client's AOV)

Payment after Day 60: Client pays **15% of estimated revenue saved**

Bound	Amount	Purpose
Floor	₹999/month	Minimum — not pocket money, covers service cost
Sweet spot	₹2,000–4,000/month	Where most small beauty stores land
Ceiling	₹4,999/month	Beyond this → Scale tier flat pricing

No results = No bill. If the data shows no meaningful return reduction, client owes nothing.

Example Calculation (Shown to Client at Day 60)

Recommendations served:	200	
Customers who acted on it:	80	(40% conversion)
Estimated returns avoided:	24	(30% would have returned)
Client AOV:	₹800	
Estimated revenue saved:	₹19,200	
ShadeMate fee (15%):	₹2,880/month	

"You see exactly how the number is calculated. No black box. No arbitrary invoice."

Why This Works Better Than Flat Pricing

- Small store with fewer orders pays less — feels fair

- Larger store pays more — scales naturally with their growth
- Client cannot argue the bill — data justifies it automatically
- Signals confidence in the product — only businesses sure of results do this

Future Revenue Streams (Phase 2+)

- Brand collaborations with Indian beauty brands for featured recommendations
- Analytics dashboard premium tier (shade trend data per region)
- White-label licensing for larger retail chains

14. Competitive Positioning

Feature	ShadeMate	Perfect Corp	L'Oreal Modiface	Status Quo
Built for Indian skin tones	✓	X	X	X
Plug-and-play for small businesses	✓	X	X	X
Affordable (under ₹5,000/month)	✓	X	X	X
Works on any website	✓	X	X	X
Matches client's own catalogue	✓	X	X	X
No dev team needed	✓	X	X	X

Direct competitors: Perfect Corp, L'Oreal Modiface — enterprise only, expensive, Western skin tone bias

Indirect competitors: Nykaa's own shade finder — not available to other businesses

Status quo: Customer orders multiple shades, tries them, returns the rest

Our moat: Only solution targeting small/mid Indian beauty businesses, calibrated for Indian skin tone diversity, plug-and-play with zero dev requirement.

15. The Team

Name	Role	Skills	What They Own
Milind	ML Engineer	Python · FastAPI · Scikit-learn	AI skin tone detection service
Sujal	Java Backend	Spring Boot · SQLite · Twilio	Backend API + data layer

Name	Role	Skills	What They Own
Shreya	Flutter Dev & Security	Flutter · Firebase · FCM	Mobile demo app + security
Uvaish	React Developer	React.js · Tailwind · Vercel	JS widget + merchant dashboard
Nikul	UI/UX Designer	Figma · App & website design	Widget UI + dashboard UX
Harshil	UI/UX Designer	Figma · Marketing assets	Pitch deck + marketing design
Sahil	Graphic Designer	Logo · Illustrations · Brand	Brand identity + visuals

Technical Execution Reality

- **Core builders:** Milind (AI service) + Uvaish (widget + dashboard) + Shreya (mobile demo)
- **AI-assisted development** used across all components to accelerate build
- Design fully covered by Nikul, Harshil, Sahil — 3 dedicated people is a strength, not overhead

16. Go-to-Market Plan

Headline: Ahmedabad pilot → Gujarat → Pan-India

Phase 1 · Month 1-3 · AHMEDABAD PILOT

How we get clients:

- **Instagram discovery + physical walk-in** — identify Ahmedabad beauty brands with own website, walk in and open with the business case directly: *"Your customers are ordering multiple shades and returning the rest. We stop that. Free for 3 months, we handle full setup."*
- **Zero-friction free pilot** — full onboarding handled by us, no technical effort from the client. Entry point is the ROI argument, not a favour
- **Direct ROI opener** — every conversation leads with money: *"If we prevent 10 returns a month at ₹800 each, that's ₹8,000 saved. This costs you nothing for 3 months."*

Pricing model for Phase 1:

- **Results-first pricing** — zero upfront cost
- Widget runs for 60 days, we collect real recommendation and return data
- At Day 60 we show the client exactly what was saved — in rupees
- They pay 15% of that saving (floor ₹999, ceiling ₹4,999)

- No results = no bill. Zero risk on their side
- Pitch line: *"We don't ask you to trust us. We prove it in 60 days — then show you the number."*

Target: 5 results-first clients in Ahmedabad **Est. Revenue after Day 60:** 3 of 5 convert at avg ₹2,500 = **₹7,500/month MRR** **Est. CAC:** ₹0 (outreach only, no paid spend)

Phase 2 • Month 4-9 • GUJARAT GROWTH

How we get clients:

- **Case study as sales weapon** — after 30 days live, document real numbers from pilot clients (recommendations served, return reduction, repeat purchases). Every Gujarat outreach now opens with data, not a pitch
- **WhatsApp D2C seller communities** — Gujarat has active WhatsApp groups for local beauty and D2C brand owners. Share case study with a direct free trial offer — peer validation in a trusted channel
- **ROI-led cold email** — subject line: *"We reduced shade returns for a Gujarat beauty brand — here's how."* Targets Surat, Vadodara, Rajkot beauty businesses with own websites

Target: 50 paid clients across Gujarat **Est. Revenue:** 50 clients × avg ₹2,500 = **₹1.25L/month MRR**

Phase 3 • Month 10+ • PAN-INDIA SCALE

How we get clients:

- **Shopify India partner referral program** — partner with Shopify consultants who already manage small beauty brand stores across India. They refer clients, earn ₹500 per paid signup. Warm leads, zero outreach cost per client
- **Content marketing targeting beauty business owners** — weekly Instagram and LinkedIn posts: *"Why your foundation customers keep returning orders", "One button that stopped shade returns for a Gujarat brand."* Builds inbound pipeline over time
- **SaaS discovery platforms** — list on Product Hunt and IndieHackers where small business owners actively search for affordable tools. Free listing, qualified inbound

Target: 200 paid clients across India **Est. Revenue:** 200 × ₹2,500 avg = **₹5L/month MRR (~₹60L annual)**

17. Demo Plan (For Pitch)

What will be shown live:

1. A mock Indian beauty e-commerce store (Shopify development store — free)
2. "Find Your Shade" widget visible on a foundation product page
3. Customer uploads selfie → lighting check runs
4. AI detects skin tone → matches against store catalogue
5. Top 3 shade recommendations shown with "Best Pick" label and color swatches
6. Customer clicks → product added to cart directly

This demo is end-to-end real — not a mockup.

18. What Is NOT in Phase 1

To be framed as **Phase 2 roadmap** in the pitch:

- Clothing color recommendations
 - Occasion-based filtering
 - Combination (outfit + makeup) matching
 - AR virtual face try-on
 - Native Shopify App Store listing
 - WooCommerce plugin (Phase 2)
 - Predictive color extraction model (trained on accumulated data)
 - Regional fashion suggestions
 - AI stylist chat
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19. Key Pitch Lines

- *"One script tag. Five minutes. Shade-bracketing returns stop."*
- *"We intercept the return before it happens — at the moment of purchase decision."*
- *"Built for Indian skin diversity — fair to deep brown, across all lighting conditions."*
- *"Not built for Nykaa. Built for the 35,000 brands that can't afford to be Nykaa."*
- *"Every client that onboards makes our model smarter — our data moat grows with our customer base."*
- *"94% of marketers say personalization drives sales. We bring that personalization to businesses that couldn't afford it before."*

20. Confirmed Slide Structure (11 Slides)

Follows the TOPS Venture Lab 9-section guide exactly.

One slide per guide section. No bloat.

Slide	Title	Guide Section	Key Content
1	Cover	—	Product name, tagline, team
2	The Problem	Section 1	Shade-bracketing story, cost to small businesses, why large players solved it but small can't
3	The Solution	Section 2	Widget flow diagram, script tag, colour swatch output
4	Target Audience	Section 3	Paying customer vs end user — two clear boxes
5	Market Opportunity	Section 4	\$28B market, 39% growth, 35,000 stores, Year 1→3 revenue
6	Unique Value Proposition	Section 5	UVP formula + competitive table
7	Business Model	Section 6	Results-first pricing, 15% formula, floor/ceiling, example calc
8	Competition	Section 7	Direct, indirect, status quo — honest comparison
9	Go-to-Market	Section 8	3-phase cards: Ahmedabad → Gujarat → Pan-India
10	The Team	Section 9	7-person grid, roles, what each owns
11	Closing	—	One strong line, current status, next step

Rules for Building the Deck

- No student framing anywhere — pitch as a business
- Never cite a specific beauty return rate — use shade-bracketing behavior framing
- Pricing is results-first — 15% of savings, floor ₹999, ceiling ₹4,999
- Target market is ALL small/mid beauty stores with own website — not Shopify only
- Competition slide must include Status Quo as a column
- GTM phases must show distinct acquisition mechanism per phase — not generic outreach
- Output shown to customer is colour swatch comparison card — NOT AR try-on

- AR try-on is Phase 2 roadmap only
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*Document Version 1.3 — Updated with confirmed market numbers and broader target definition.
All architecture decisions confirmed through discussion.
Do not share externally before pitch.*